Lensless Dragonfly-sized Robots: Minimal Depth Perception for Micro-Quad Autonomy

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***Abstract*—Micro aerial robots require compact and lightweight vision systems to navigate cluttered environments because traditional lens-based cameras are too bulky and power-intensive. To overcome these limitations, we developed a lensless imaging system that enables depth perception using a custom-fabricated diffractive optical element (DOE). During A Term, we investigated past approaches to lensless imaging and related optical systems, including diffuser-based and PiCam implementations, and finally, recreating and building upon optical fabrication to U-Net testing for depth estimation.**

I. Introduction

Safe navigation in cluttered environments such as dense forests, urban infrastructure, and disaster environments is crucial [1]. Applications for autonomous aerial robots include search and rescue [2][3], disaster response, environmental monitoring, infrastructure inspection, and defense [4]. In such scenarios, autonomous aerial robots must be able to navigate through narrow gaps and avoid obstacles with minimal clearance while maintaining flight stability. It is not only a technical challenge for aerial robots to operate safely in complex environments, but a practical necessity when operating in environments that pose high risks to human operators.

Conventional aerial robots are often limited by their size, weight, and power limitations, which in turn reduce their speed, making them less effective in confined or unpredictable spaces. Alternatively, micro aerial robots are smaller, lighter, and highly maneuverable, with the ability to maneuver in places that would be inaccessible with traditional aerial robots. However, due to the strict constraints on payload and power, it requires the use of lightweight, efficient sensing and navigation systems.

Vision-based sensing provides a promising solution for enabling autonomous flight in these compact platforms. Many existing systems rely on traditional lens-based cameras, which present valuable visual cues for navigation. While these cameras are effective, they introduce several drawbacks: they are bulky, add weight, increase power consumption, and are expensive. These limitations reduce the feasibility of integrating conventional imaging systems into ultra-compact aerial robots.

Lensless imaging has emerged as a compelling alternative to address these challenges. These systems offer ultra-thin, lightweight imaging systems with optical elements and eliminate bulky optics, making them suitable for micro aerial platforms [5]. In addition to minimizing payload, lensless architectures can support computational depth estimation within a single, compact device, enhancing navigation capability without increasing hardware complexity. This allows for a robust and nimble micro aerial robot to navigate dense and cluttered environments.

In the current active field of research, various lensless imaging approaches have been explored, such as amplitude masks, coded apertures, and phase masks. Amplitude masks transmit incoming light and modulate the incoming light intensity, allowing image reconstruction through computation, but often at the cost of resolution [6]. Coded apertures improve spatial resolution through structured patterns. Finally, phase masks modulate the phase rather than the intensity of light, offering better support for depth estimation tasks [7]. Among these, phase-mask-based systems have shown potential for compact, depth-capable vision because of their ability to encode three-dimensional scene information into a 2D capture. This makes them well-suited for aerial robotic applications where both size and computational efficiency are essential.

Building on this foundation, we propose a lensless imaging system that uses a phase mask for depth estimation in aerial robotic navigation. Our system integrates a custom-manufactured phase mask with an image sensor to capture optical information directly, without the use of traditional lenses. The optical blur resulting from is then processed through our computational imaging pipeline, which reconstructs depth information from the raw sensor data and provides spatially meaningful visual cues for navigation (Full Goal Of The Project).

Our experimental work during A-Term expands upon prior research in the field of computational imaging and lensless vision. DiffuserCam experiments showed the potential of diffuser-based lensless imaging, achieving a compact optical design at the cost of reduced spatial resolution [8]. The Lensless PiCam experiments focused on developing reconstruction algorithms with depth estimation accuracy [9].

In addition, we evaluated a diffractive optical element (DOE) in-depth estimation by recreating the PhaseCam3D in fabricating a custom phase mask [10]. We mounted this DOE on an ArduCam and tested the system alongside an Intel RealSense camera, which served as ground truth for depth comparison. We, then, captured the data to evaluate the reconstruction quality and depth estimation performance using a U-Net inspired by the Deep Optics Monocular Depth Estimation framework[11]. These experiments collectively demonstrate our approach to integrating optical design, DOE fabrication and manufacturing, and neural network-based reconstruction for lensless imaging in aerial robotics.

II. Related Works

A. DiffuserCam

DiffuserCam demonstrated that depth-dependent PSF variation is enough to perform 3D reconstruction, even without explicit chromatic dispersion or learned optimization [8]. However, understanding how DiffuserCam worked was valuable in grasping how unstructured optics could encode 3D information without utilizing a traditional lens. The paper emphasizes how random scattering materials can still encode depth information as long as their point spread functions (PSFs) vary sufficiently in space. This insight was invaluable in formulating our initial experiments with diffusers before completely switching to optimized DOEs.

We require our phase mask to produce PSFs that vary in a predictable manner with depth. The experiment performed with DiffuserCam suggests that this approach holds promise, as it models the physical optics sample by using a random phase mask with varying random diffusing patterns.

B. Design and single-shot fabrication of lensless cameras with arbitrary point spread functions

This paper explores lens-free imaging research and development, starting with early approaches based on coded apertures and weak diffuse reflection. Thin masks replaced bulky lenses, which made it possible for thinner, low-cost cameras to be made. Limiting masks to just phase-modulation is efficient in collecting light and having PSFs that are shift-invariant, allowing for robust reconstruction. This offers a foundation for understanding how PSFs can represent 3D images, as the aim is to use a designed PSF to decode 3D depth information from a single shot.

The authors use a single-shot gray-scale maskless photolithography process to fabricate the phase masks employing a digital micromirror device (DMD) that is used to project a spatially varying intensity of UV light onto a layer of photoresist. This process is high-throughput for phase mask fabrication as it is simple and cost-effective to synthesize photonic optical elements over a relatively large area, significantly superior to two-photon polymerization, which can be very precise but slow and less scalable.

Once the masks have been fabricated, they are directly aligned into a clear form-fitted top of a CMOS sensor, using a suitable spacer to fix the focal distance. The camera captures one raw intensity image, which encodes the 3D scene information by spatially varying PSF. Subsequently, a high-quality estimate of the original scene can be retrieved by computationally reconstructing the captured image using an alternating direction method of multipliers (ADMM) optimization problem (linear) using total variation (TV) regularization. The authors demonstrated successful reconstruction across a number of different PSF types .

The method provides an alternative for custom phase mask design and fabrication by employing a phase-retrieval algorithm with smoothness constraints to generate phase profiles suitable for high-throughput grayscale lithography. The generated masks are designed to produce contour patterns that provide raw 2D data when shaken with a single image sensor that will reconstruct high-quality 3D scenes via computational reconstruction using ADMM-based regularization.

C. PhlatCam - Designed Phase-Mask Based Thin Lensless Camera

PhlatCam [7] is a lensless camera that replaces a traditional lens with a custom-designed phase mask mounted less than 2 mm from an imaging sensor. Instead of forming an optical image, the mask diffracts incoming light into a designed PSF, and the resulting encoded 2D capture is computationally inverted to recover the scene. The mask is fabricated using two-photon lithography, and the system is calibrated with depth-dependent PSFs to enable high-quality reconstructions.

A key innovation is the Near-field Phase Retrieval (NfPR) algorithm, which designs the mask to produce the desired PSF at a specified thickness by modeling near-field Fresnel propagation. The encoded sensor data is then decoded using deconvolution and optimization-based reconstruction algorithms. In addition, a contour-based PSF, generated from Canny edges of Perlin noise, is introduced using signal processing heuristics to achieve a wide and flat frequency spectrum, high contrast, and spatial sparsity. These properties improve information retention and allow for robust, high-quality image reconstruction using both fast linear and advanced nonlinear techniques.

PhlatCam's design achieves high-fidelity photography (2D imaging up to 3m range), microscopy (close-up fluorescence imaging), computational refocusing, and 3D depth imaging within 0–30 mm range [7].

This paper is useful for our proposed system as it provides the foundations for the optical design and computational framework for building a custom lensless imaging system, especially for depth estimation. PhlatCam describes how PSFs are depth-dependent, allowing us to learn to predict how scene depth can translate into measurable optical differences. Finally, their algorithm can help guide us for our DOE or phase mask optimization.

D. PhaseCam3D

PhaseCam3D [10] demonstrated a method for end-to-end learned optics for metric depth estimation. Importantly, this approach was not lensless; the system relied on conventional refractive optics, and the primary contribution of the paper was a simulation framework rather than a robust, real-world deployment.

Wu et al. presented a technique for simulating defocus blur caused by an arbitrary phase mask placed at the focal plane of a camera. Using this forward model, they could generate synthetic data to jointly train both a depth estimation network and the phase mask geometry itself. Through this end-to-end optimization pipeline, a custom phase mask could be designed for a target depth range given the optical properties of the imaging system.

To enable this optimization, Wu et al. proposed a loss function for optical design derived from prior work in 3D microscopy [12], where a statically optimal phase mask was computed using a Fisher Information Function.

$$\begin{aligned} I_{ij}(\theta) &= \sum_{t=1}^{N_p} W \left(\frac{\partial \mathrm{PSF}_\theta(t)}{\partial \theta_i} \right) \left(\frac{\partial \mathrm{PSF}_\theta(t)}{\partial \theta_j} \right), \\ W &= \frac{1}{\mathrm{PSF}_\theta(t) + \beta} \end{aligned} \tag{1}$$

This formulation defines the Fisher Information Matrix $I_{ij}(\theta)$, which measures the sensitivity of the (PSF) across different camera states, θ_i. The term W weights each pixel inversely proportional to its noise level. The Cramér–Rao Lower Bound (CRLB) is then obtained from the inverse of the Fisher matrix, as shown in (2). The diagonal elements of $I(\theta)^{-1}$ represent the variance in state estimates (x, y, z) that the PSF induces.

$$\mathrm{CRLB}i \equiv \sigma_i^2 = \left[I(\theta)^{-1} \right] ii. \tag{2}$$

By summing the square roots of these variances across spatial and chromatic dimensions, we obtain a single scalar loss that quantifies how the PSF behaves across depth and color:

$$L_{\mathrm{CRLB}} = \sum_{i \in x,y,z} \sum_{z \in \mathcal{Z}} \sum_{c \in R,G,B} \sqrt{\mathrm{CRLB}_i(z,c)}. \tag{3}$$

This loss function can be applied independently of PhaseCam3D's simulation framework to optimize a phase mask using only Fourier optics to model the propagation of a point source through a thin lens. This standalone optimization produces a theoretically optimal mask for a sparse scene with a single emitter [12]. Wu et al. used this CRLB-based approach to generate an analytically optimized reference mask (Fisher mask), which they compared against their end-to-end learned phase masks.

Crucially, the authors found that the loss surface for end-to-end optical optimization is highly non-convex, and the phase mask geometry often converges to local minima with inferior performance compared to their Fisher Mask. To address this, they initialized the learned optics model with the CRLB-optimized mask before continuing end-to-end training, allowing convergence toward a globally optimal design.

Finally, the CRLB-initialized, learned phase mask was fabricated and experimentally evaluated in indoor scenes against the analytically optimized mask. Although the learned mask substantially outperformed the CRLB-optimized mask in simulation, in real-world experiments over a depth range of 0.4 m to 0.9 m, the PhaseCam3D learned mask achieved a depth estimation error of 0.0125 m RMS, while the Fisher-optimized mask achieved a higher, but still acuate, RMS error of 0.0308 m.

These results show that while there may be much performance to be gained by using an end-to-end learned phasemask, the Fisher-optimized mask still performs very well and is worth investigating. Its underlying math is much simpler and therefore more transferable to the real world, where we are interested in deploying our system. Using these results, we began the manufacturing process for our own Fisher-Optimized mask.

E. CodedVO

CodedVO [13] was developed by the same research group behind [14] and can be seen as a natural continuation of that work. Shah et al. use the learned phase mask from PhaseCam3D [10] and deploy it on a simulated 1-inch sensor camera with an f/1.8 lens. The nonlinear blur model from [14] is used to generate synthetic training data, which is then augmented to incorporate the effects of the phase mask. This augmented dataset is used to train a simple U-Net to predict depth from single indoor images with distances up to 6 m.

Their training process employs a depth-weighted loss function that emphasizes accuracy for closer objects by exponentially weighting the L2 loss according to the ground-truth depth value, as defined in (4):

$$\mathcal{L}dw = \frac{1}{N} \sum_{i=1}^{N} w_{\hat{D}_i} \cdot (\bar{D}_i - \hat{D}i)^2, \qquad w\hat{D}_i = \alpha^{-\beta \hat{D}_i}. \tag{4}$$

This weighting scheme penalizes errors more heavily for nearby objects, which are generally more relevant for navigation and occlusion reasoning.

Shah et al. achieved such strong metric depth estimation results with their simple model that the predicted depth maps could be passed directly into existing visual odometry pipelines, producing excellent performance.

For our purposes, CodedVO is particularly insightful because it demonstrates that the training method used for the phase mask generation does not constrain its applicability. The phase mask from PhaseCam3D was trained using a simplified, linear blur model that did not accurately capture occlusion-related effects. Yet, it could still be seamlessly integrated into a nonlinear, occlusion-aware model and achieve very strong results. This shows a valuable level of cross-model generalization for phasemasks: the learned optics remained effective even when the underlying imaging physics became more complex.

CodedVO's results clearly illustrate how powerful coded apertures can be for embedding depth information in a form that is computationally accessible. However, their goals and ours diverge. While Shah et al. successfully demonstrated metric depth estimation up to 6 m, our objective is instead to capture relative depth sufficient for obstacle avoidance rather than precise reconstruction. If they are truly able to achieve 6m metric depth with a simple U-Net, there is a strong indication that we can achieve around 3m relative depth with an even simpler architecture. On the other hand, since their model was never deployed in a real-world optical system, their strong simulation results should be interpreted with caution.

F. Deep Optics for Monocular Depth Estimation and 3D Object Detection

The idea of ”deep optics”[15] is one of the works that focuses on monocular depth estimation, and this paper is an extension of Depth for Defocus [14]. The authors proposed that coded defocus blur could be another depth cue besides classic pictorial depth cues. The optical encoder consisted of a fixed aberrant lens or optimizable freeform lenses, while a U-Net acted as the electronic decoder. The paper evaluates several optical coding strategies:

- Defocus only: Objects in a standard lens that are at the same distance from the focus point will be equally blurry, which can hinder depth perception.
- Astigmatism: Indicates asymmetric blurring of the image by using horizontal and vertical lines.
- Chromatic aberration: Indicates blurring of the image based on the lens blur from a singlet lens, where different wavelengths are focused at different distances.
- Optimized freeform mask: Fully learnable phase profile parameterized with 36 Zernike polynomials (Zernike polynomials are a set of mathematical functions used to represent and model complex wavefront shapes or optical aberrations on a circular aperture).

Their differentiable image formation method was originally derived from wave optics, where the light was assumed to propagate through only the phase mask and a thin lens, which generates depth-dependent PSFs. Those PSFs were then convoluted with the image layers across depth to create realistic training data. The optimization iteratively updated both optical parameters and network weights, while minimizing the loss in depth prediction error on datasets including NYU Depth v2 and KITTI.

The prototype depth estimation system had several limitations. (1) It had a limited field and a spatially varying PSF. (2) Used pseudo-ground truth depth maps (rather than true ground truth) for the training set. (3) Used a rough layer-based image formation model and did not account for partial occlusions in the forward model, which could potentially impair the accuracy and robustness of the depth predictions.

This paper is supplemental to our approach. The key insight that simple chromatic aberrations can nearly match optimized designs suggests we could potentially use off-the-shelf singlet lenses or simple DOE designs rather than requiring complex fabrication. However, for our application, we need even more compact solutions. Their end-to-end optimization framework provides the blueprint for our training pipeline. The paper demonstrates that absolute depth estimation is achievable with coded defocus, which is critical for obstacle avoidance in autonomous flight.

G. Depth Estimation from a Single Image using Deep Learned Phase Coded Mask

They introduce a phase-coded aperture camera for estimating depth from a single image, utilizing a jointly learned phase mask and a deep neural network [16]. Their training framework jointly optimizes the phase mask design and the fully convolutional network (FCN) using backpropagation. At a high level, the mask causes tiny, wavelength-dependent, and defocus patterns that vary with distance, allowing the networks to decode the color-coded blur into quantitative metric depth maps.

One of the main takeaways from this study is the joint learning framework, which provides significant improvement in accuracy with a major computational complexity reduction. The neural network can be made much simpler than what is typically used in monocular depth architectures (e.g. ResNets, multi-scale CRF-based models), because the optical mask encodes rich depth information. The authors trained their model on a custom synthetic high resolution dataset. Results from simulated and real optical experiments demonstrated faster inference and lower error overall.

In contrast to previous studies using only simulations, this one constructed an actual optical phase mask experiment on a camera with a lens (16 mm, F/7 lens). This study demonstrates that optical coding can alleviate computation bottlenecks by embedding depth cues right into the captured image.

III. Manufacturing

A. Fabrication Process for our Custom-Designed Phase Mask

The fabrication process involves using 2PP to fabricate a PUA mask that can be tuned and made with submicron precision [17]. In this process, a laser is focused tightly to cause localized polymerization within the photosensitive resin, enabling 3D structuring of the designed phase profile. The advantages of the 2PP method include high spatial resolution for the order of the mask’s submicron features; the single-exposure format insures repeatability for fixed designs, and 2PP masks scale up rapidly for custom DOEs and rapid prototyping.

B. Heightmap Generation and STL Conversion

Before we begin manufacturing, we had to convert our reference 2D heightmap into an STL that can be used by the Nanoscribe slicing and generating software - Describe.

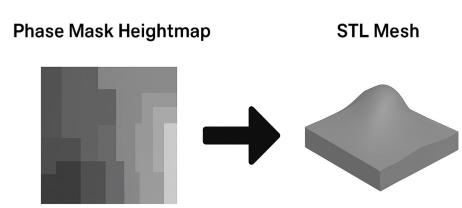

Fig. 1. Heightmap-STL Conversion

We attempted to do this by using on of our notable references, Depth for Defocus [14], and their method of converting a 2D grayscale heightmap image with customizable parameters as inputs. The heightmap (430×430 pixels, 8-bit grayscale) must be converted to STL format for Nanoscribe import. We used an open-source Python script that used dynamic subdivision [18].

C. Fabrication Technique: Two-Photon Polymerization (Nanoscribe)

We opted for the Nanoscribe Photonic Professional GT2 at WPI's Lab for Education and Application Prototypes (LEAP) facility over the grayscale lithography approach used in earlier work [19].

These are the initial design parameters we selected to experiment with, largely inspired by (1) PhaseCam3D [10], (2) Depth from Defocus [14], and (3) Design and single-shot [20] who which used similar diffractive mask fabrication methods and Nanoscribe-based processes.

TABLE I
INITIAL PHASE MASK DESIGN PARAMETERS

Parameter	**Specification**
Footprint (X × Y)	1.5 mm × 1.5 mm
Maximum Height (Z)	2.1 μm
Height Profile	Radially symmetric
Layer Count	$\sim$10 layers
Wavelength (λ)	589 nm

This process writes 3D microstructures directly into a photoresist using a focused laser, offering sub-micron precision. The workflow consisted of substrate preparation, digital model slicing, laser writing, and post-processing [17].

1) Substrate Preparation: Started with a 25×25 mm glass slide (1 mm thick). Cleaned with acetone, isopropanol (IPA), and deionized water. No additional coating was needed since two-photon polymerization bonds directly to glass.

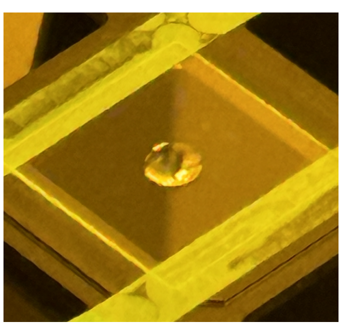

Fig. 2. Resin on substrate

2) STL Import and Slicing: Imported the STL design file into Nanoscribe Describe software.
We opted for a 25× objective lens (NA 0.8) for a 1.5×1.5 mm area, or 63× (NA 1.4) for finer features. Chose a layer thickness of 0.2 µm and a hatching distance of 0.3 µm (lateral spacing between scan lines). The slicing process generated roughly 10 layers for a total structure height of 2.1 µm.

Fig. 3. 1.5mm X 1.5mm DOE

3) Two-Photon Polymerization: Loaded the glass substrate into the Nanoscribe chamber. Apply a 5–10 µL droplet of photoresist (IP-S or IP-Dip). Operated in dip-in mode, immersing the objective lens in the resist.

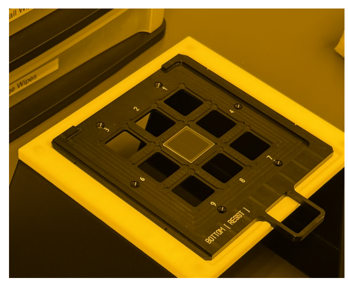

Fig. 4. Substrate placed on carrier

Write time was approximately 122 minutes, depending on complexity.

4) Development: Immersed the sample in PGMEA (Propylene glycol monomethyl ether acetate) developer for 10 minutes to remove unexposed resist. Then rinsed with IPA and dried with nitrogen.

Fig. 5. PGMEA Diluter for Cleansing DOE

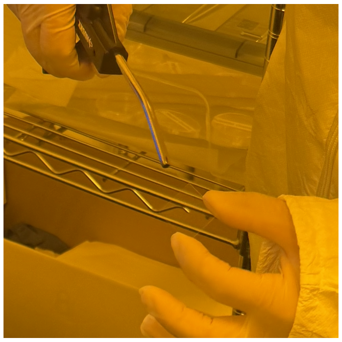

Fig. 6. Nitrogen Vacuum for Air Drying

5) Post-Processing: Inspected the finished mask under an optical microscope to verify the 2.1 μm height and surface smoothness.

D. Validating DOE Fabrication Quality - Zygo Microscopy

After fabricating the DOE via Nanoscribe two-photon polymerization, the most important step in validation is verifying the optical profile to ensure that the fabricated DOE corresponds to the designed height map. This is critical as even minor variations (±100nm) can significantly reduce the quality of the PSF and depth discrimination. We utilized Zygo white-light for non-contact, sub-nanometer resolution surface characterization.

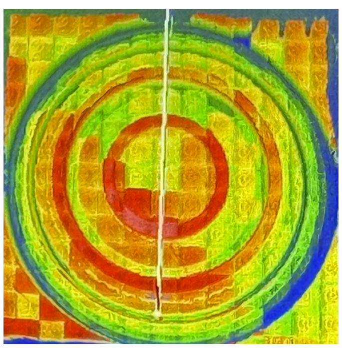

Fig. 7. Height Variation Map

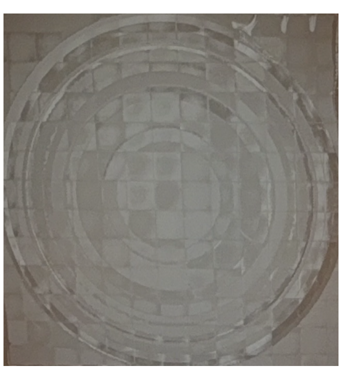

Fig. 8. Hatching Grid

The microscopy analysis further helped us determine that our DOE matched what we expect to see in terms of the physical attributes and assured that there was no immediate, clear deformities. The infrared measurement Fig. 7 tells us that there is variation in the contours and the layer heights, which is what we expect to see in our DOE.

IV. Experiments

A. Diffuser Cam

DiffuserCam is a lensless 3D imaging system that replaces traditional optics with a diffuser placed directly in front of an image sensor. The setup involved a simple piece of Scotch tape, acting as a diffuser, placed over a coded aperture, with a focal distance of approximately 3.5 mm from the image sensor. The calibration was done by using a pinhole point-source LED in a dark room to avoid stray light, which resulted in a PSF. Each distance in the scene provided a PSF that encodes spatial and depth information. Though the PSF is depth-dependent, during evaluation, it was only dependent up to a range of 10 cm. The objects tested were high-contrast black-and-white images displayed on an illuminated screen at 40 cm, matching the PSF calibration distance.

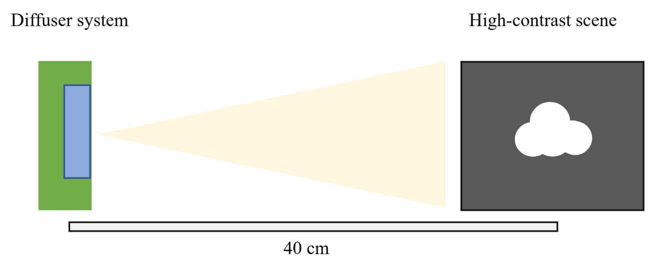

Fig. 9. Diffuser imaging system physical setup

Image reconstruction is formulated as an inverse optimization, meaning they must recover a 3D scene, x, from a 2D sensor measurement b given a system model H, i.e.:

$$\min_{x} \; \|Hx - b\|_2^2 + \lambda R(x)$$

where R(x) is a regularization term that promotes sparsity or smoothness. To solve this, we tested several iterative optimization algorithms such as Gradient descent, Nesterov, Fast Iterative Shrinkage-Thresholding Algorithm (FISTA), and Alternating Direction Method of Multipliers (ADMM).

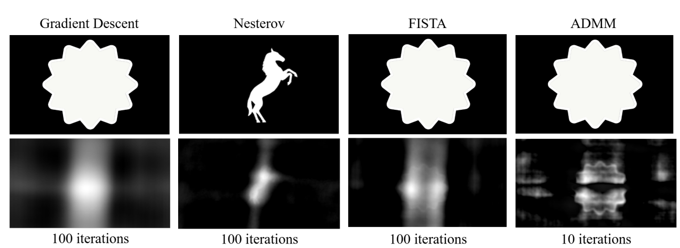

Fig. 10. The results of image reconstruction optimizers and algorithms

Gradient Descent updates the estimate by moving in the direction opposite to the gradient of the loss function iteratively. The optimizer is simple to implement, but converges slowly and is highly sensitive to the choice of step size. The Nesterov Accelerated Gradient optimizer improves upon the former method by incorporating a momentum term that anticipates the next step. This converges faster and reduces oscillations near the optimum. FISTA combines Nesterov's acceleration and a shrinkage operator, which handles non-smooth regularization terms or penalties. FISTA is considerably faster to converge than the past two methods. The best results were achieved using ADMM, an optimization framework that splits the objective into chunks of subproblems for data fidelity and regularization. ADMM manages large-scale, ill-posed, and constrained problems, providing superior stability and reconstruction accuracy at the cost of higher per-iteration computation time. Overall, while FISTA offered the best balance of speed and simplicity for 2D reconstruction, ADMM gives the best reconstructions for full 3D.

B. Lensless PiCam

Based on our relative success with the DiffuserCam repository, our next step was to evaluate a system with more features, Lensless PiCam [9]. This repository provided functionality for RGB image reconstruction and dataset collection features to train a model. We used a Raspberry Pi 4 to display an image on a monitor in order to collect large amounts of training data autonomously. The result of a reconstructed is shown in Fig. 11, which was reconstructed using 2500 ADMM iterations taking about 2 seconds on an RTX 3060 Max Q. The result was worse than the DiffuserCam results. This is likely due to a lack of tuning in the network and its complex RGB image reconstruction algorithm.

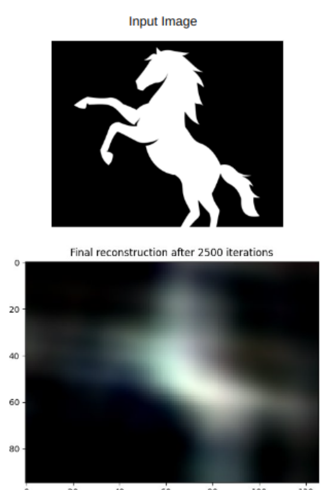

Fig. 11. Image reconstruction of horse with LenslessPiCam

Ultimately, we abandoned this framework after determining that diffusers would not fit our requirement for 3D imaging.

C. Diffuser Testing

The diffuser testing was done to evaluate the depth-dependent PSFs that [7] uses. We used a point-source LED to retrieve PSFs at different distances.

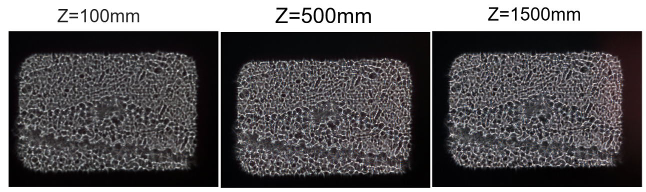

Fig. 12. PSFs at different distances with scotch tape as the diffuser

We were unable to see the changes in the overall shape of the PSF. The expectation was that the rectangle of the PSF at Z = 100 mm would be smaller than when Z = 1500 mm. Furthermore, we decided to test different materials testing including the off-the-shelf Diffuser sheet, LCD Diffuser Sheet, Kapton Tape, and DigiKey ESD Safe Bag, for depth-dependence.

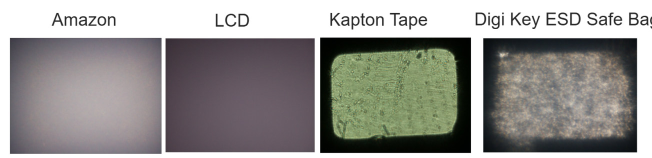

Fig. 13. The PSFs of different diffuser materials

The PSFs shown in Fig. 13 were worse than the scotch tape used in 12, except for the DigiKey ESD Safe Bag, which can be seen a bit more clearly, so we evaluated it further by checking its PSF at different depths.

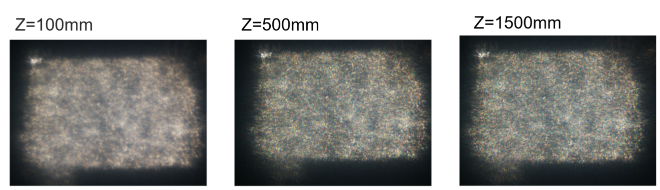

Fig. 14. Diffuser PSFs at different depths

In three images, Fig. 14 were similar in size regardless of the change in distance. The cue that changed was only the sharp contours when it's further from the image sensor. In short, this was not depth-dependent.

In conclusion, the diffuser materials that were evaluated were not depth-dependent, so we cannot get much information from them to perform 3D image reconstruction or get depth estimation.

D. Recreating PhaseCam3D Network

With the code for this paper publicly available, the group investigated it thoroughly, training our own phase masks. Compared to the complexity with which the math to model the optics was presented in the paper, the code was surprisingly simple and easy to follow. It was incredibly helpful to see the numerical methods that the authors used to transfer the math from the paper into code. Most notably, the number of application-specific parameters was very limited, meaning it was very easy to adapt the design to fit our optics. Unfortunately, the code was missing the Cramér-Rao Lower Bound loss function and the proper framework to initialize the CRLB optimized mask, so all the phase masks we generated were likely local minima and not correctly optimized.

We implemented our own Cramér-Rao Lower Bound loss, but we were limited on time. Finally, we will want to our optics to be lensless, which will change how we propagate the PSF, meaning we have to modify some of the math. According to [7], we simply replace the Fourier Transform with near-field Fresnel Propagation, but we need to see how to transfer that math into code.

E. Testing DOE

We fabricated a custom-designed DOE to evaluate depth-dependent modulation. The DOE was tested to analyze how the diffraction pattern changes with varying object distances. In 15, the physical setup includes the DOE-fabricated module mounted on an ArduCam with Jetson Orin Nano.

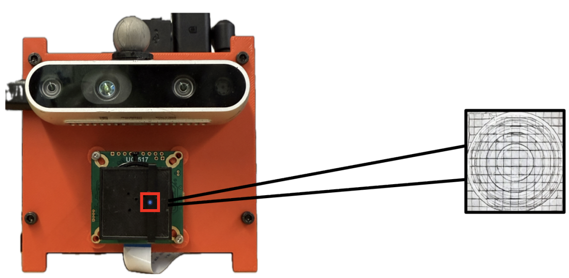

Fig. 15. Custom DOE Integration on Arducam CMOS

To retrieve the PSFs, we used a single point source and collected data at different depths. The observed depth-dependent in Fig 16, variations confirm the potential of DOE for depth-sensing and computational imaging. Preliminary results demonstrate variations in the captured diffraction patterns as a function of depth, confirming that DOE exhibits depth-dependent optical behavior.

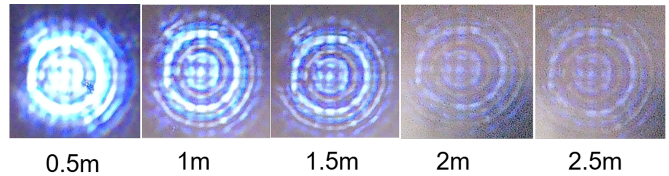

Fig. 16. PSFs of Custom DOE at Different Depths

For 0.5 m, the contours were closed together in distance, but stretched gradually as the point source was pointed at 2.5 m. At each distance, another key observable characteristic was intensity. The closer the distance, the higher the intensity in each contour. The gradual changes in contour shape and intensity distribution correspond to the dilation of the optical wavefront, resulting in all the observable patterns. In short, the optical wavefront was effectively changed, resulting in depth-dependency, which is a fundamental requirement for depth reconstruction this led to the next step of quantitatively testing the DOE's capability to obtain depth information.

F. Testing BiConvex Lens

Beyond the DOE, we evaluated an off-the-shelf, cheap biconvex lens to assess its potential for PSF depth-dependent modulation [15]. The lens, with a focal length of 45 mm, produced a significant magnification effect.

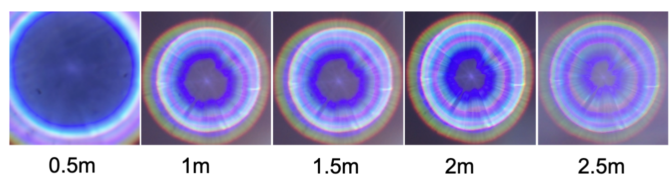

Fig. 17. PSFs of 45mm Convex Lens at Different Depth

This results in an extremely narrow field of view of approximately 7 degrees, which was impractical for further testing on depth-sensing applications. Due to these constraints, further experimentation with this optical design was discontinued.

G. U-Net with ArduCam with DOE and RealSense

The experimental setup consisted of an NVIDIA Jetson platform, a custom-fabricated DOE with a pinhole aperture mounted onto an ArduCam, and an Intel RealSense camera used for ground truth. The first step involved synchronizing the two imaging systems to ensure simultaneous frame capture. Because RealSense has a larger field of view, it was software-cropped at the center to match ArduCam.

A classic U-Net model was developed to decode the DOE images into a relative depth map. The U-Net accepted a 3-channel RGB-coded image and output a depth map and uncertainty map [21]. Finally, the loss was weighted around the center and zeroed at the borders of the image to account for the different FOV of the input and label. L2, Huber, and Smooth L1 loss were experimented with, and ultimately Smooth L1 with Beta of 0.1 was used. The hypothesis was that uncertainty would help the model converge and give valuable clues about what was causing depth estimation to break down with different phase mask designs.

In order to efficiently manage the large heterogeneous data sets, we implemented a custom PyTorch dataloader that then pre-processes images from multiple sources. The data collection had approximately 67,000 captured indoor images for testing and analysis, but many were discarded due to the misalignment of the software crop. The depth frames were removed if the percentage of depth pixels fell below our threshold, yielding only usable image pairs for model training, reducing the dataset to 3,461 sets being used to train. Both RGB (DOE ArduCam) and depth (RealSense) data underwent the same geometric transformation processes, resolution matching, and data normalization.

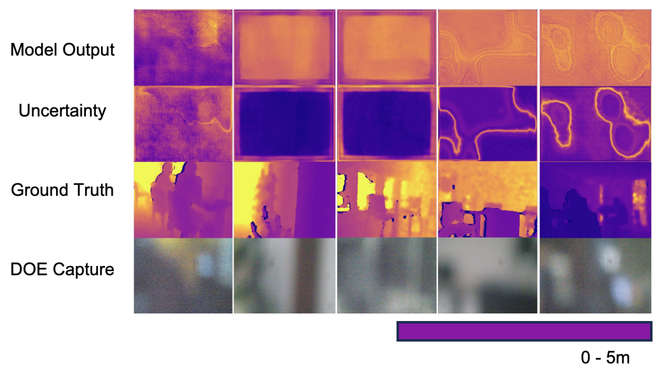

Fig. 18. Indoor Images Captured

The results in Fig. 18 showed unfavorable results where the model doesn't output images with any resemblance to the ground truth. This may have occurred because of the misalignment in cropping and an insufficient dataset. However, there was limited time to re-capture images and retrain the model. The work will be continued in B term, along with other methods.

V. Next Steps

The next stage will emphasize further developing the optical and computational parts of the apertures. This involves trying different network architectures and developing more custom phase masks, including simulated end-to-end networks that also use mathematically optimized phase masks. More complex data simulation will be developed to better reproduce the results from previous studies and train models more effectively. Hardware will also be improved through the usage of a biconvex lens with a smaller focal distance to gather data that is more accurate and reproducible. The data gathering from the RealSense and custom DOEs will be improved through the simultaneous collection of ground truth, including precise mathematical cropping and software calibration to match the fields of view between the cameras. Collectively, we hope this will produce more accurate, consistent, and generalizable results in depth estimation.

VI. Conclusion

In summary, our efforts during A-Term laid the groundwork for a compact, yet lensless imaging system, as we explored previous works to learn and recreate. This was a stepping stone to learn what worked and didn't for the future, in which we hope to fully incorporate depth perception for micro aerial robotic platforms. We initially investigated lensless imaging systems, particularly using diffuserCam and Lensless PiCam, to understand how depth information can be optically encoded without the use of lenses. Subsequently, we developed, manufactured, and tested a custom-designed DOE, employing two-photon polymerization methods with the Nanoscribe to achieve the structure, and verified it using ZYGO microscopy before pairing the DOE with an ArduCam sensor. The experimental tests confirmed that the DOE produced patterns invariant with depth. Additionally, we took a step further to implement our custom U-Net architecture. Overall, we developed the optical design, manufactured a DOE, and implemented and tested a computational depth system, which serves as a foundation for our work in B-term.

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